

A student team building an autonomous boat from the ground up.

A composite planing hull, two motor pods and a rudder, designed as one system at UW-Madison.



HOW WE MODEL IT

First-principles mixed-integer optimization

We are building the model from physics up, so every part of the boat is tied to every other. Change the hull, the battery or the speed and the effect carries through the whole boat. Component choices are integers; sizes and speed are continuous.

Hull, battery, drive, speed → Mass and balance → Resistance →
Power → Battery energy → Race time

choose the one boat that passes
every margin, fastest first

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minimize  $\sum T_i \cdot x_i$   
s.t.  $\sum x_i = 1$   
 $\sum m_{ki} \cdot x_i \geq 0 \quad \forall k$   
each  $x_i$  is 0 or 1
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ENGINEERS FROM ACROSS UW-MADISON

Each segment of the boat is covered, and members have worked at Tesla, Northrop Grumman, Xcel Energy and Milwaukee Tool.

Industrial

Optimization, composites
and manufacturing

Mechanical

Structures, CAD and
fabrication

Electrical

Power, battery and high
voltage

Computer

Embedded, firmware and
software

OUR APPROACH

Hull

Plane well and stay light.

- Hybrid planing composite hull
- Shape tuned by optimization
- Resistance checked in CFD first

Electric

Every watt-hour to speed.

- High-voltage battery storage
- Motor pods sized by the model
- Differential thrust and rudder

Autonomous

Find the way, unaided.

- GPS, IMU and LIDAR sensing
- Firmware protects the battery
- Software plans the route

TIMELINE

Oct 25
Hull design

Nov 15
Electrical design

Nov 25
Hull built

Jan 25
First drive

Feb 15
First fast drive

Mar 15
Tuning and testing

Sponsor the build

We are looking for funds, parts and advisors.

LEARN MORE

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CONTACT

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